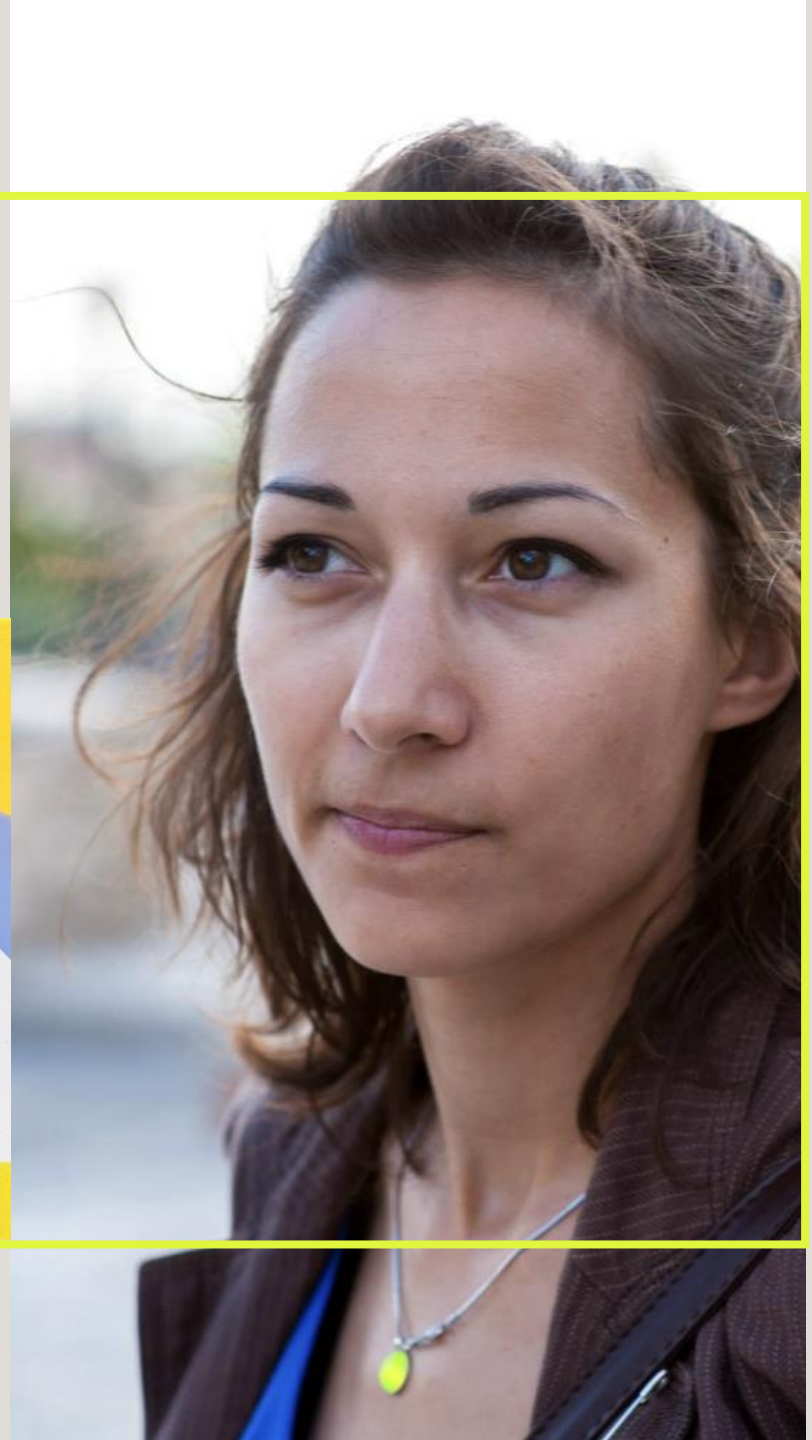
BANKING SYSTEM APPLICATION – SECURE, INTERACTIVE, AND SCALABLE

(Stallmann, 2019)

INTRODUCTION TO SOFTWARE ENGINEERING



Assessment 3

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Course: Introduction to Software Engineering

INTRODUCTION TO THE PROJECT

Objective:

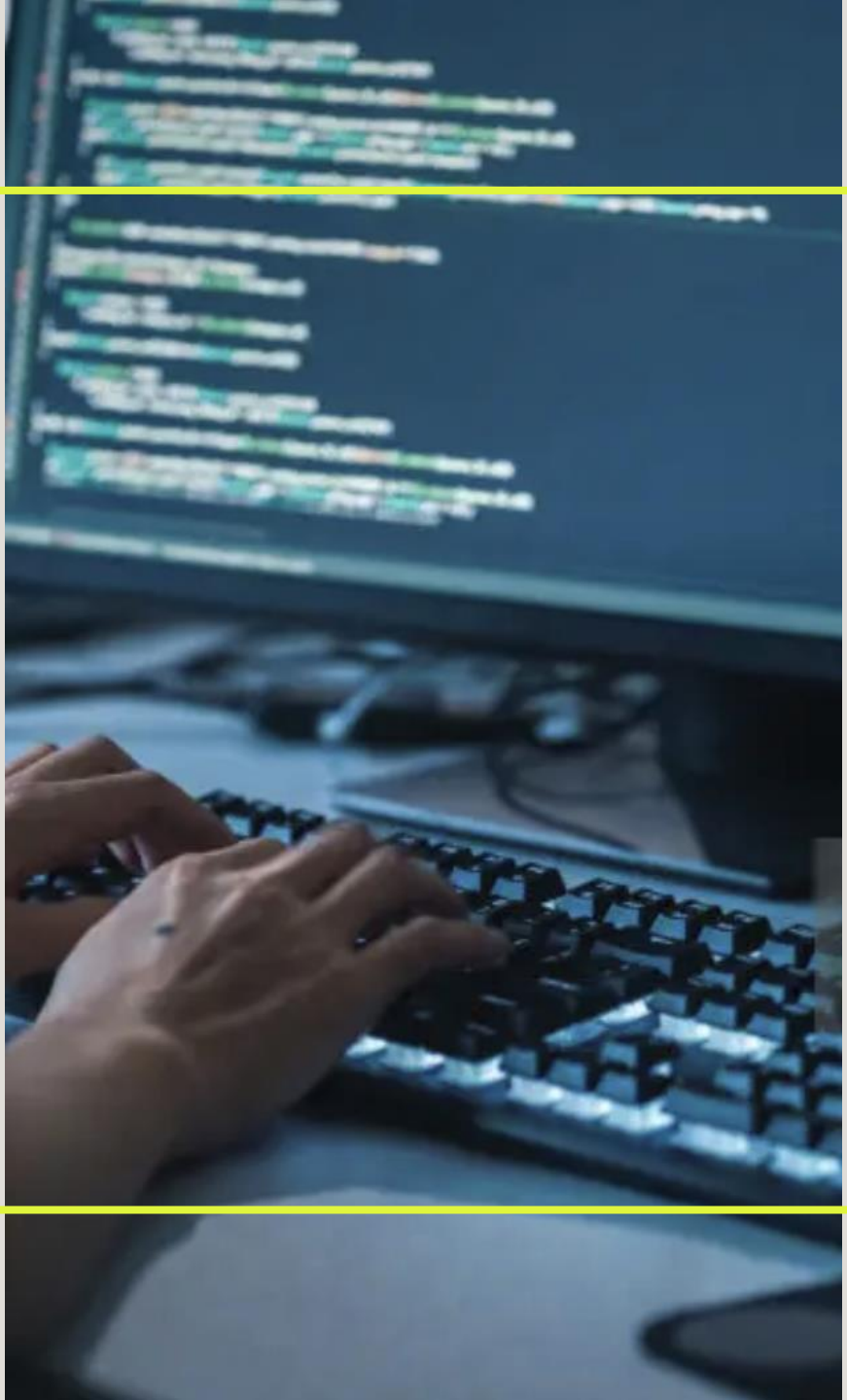
- Develop a secure and interactive banking application.

Modules:

- Signup
- Login
- Deposit
- Withdraw

Key requirements:

- Validations
- Error handling
- Secure user interactions



(*Close-up Focus on Person's Hands Typing on the Desktop Computer...*, 2021)

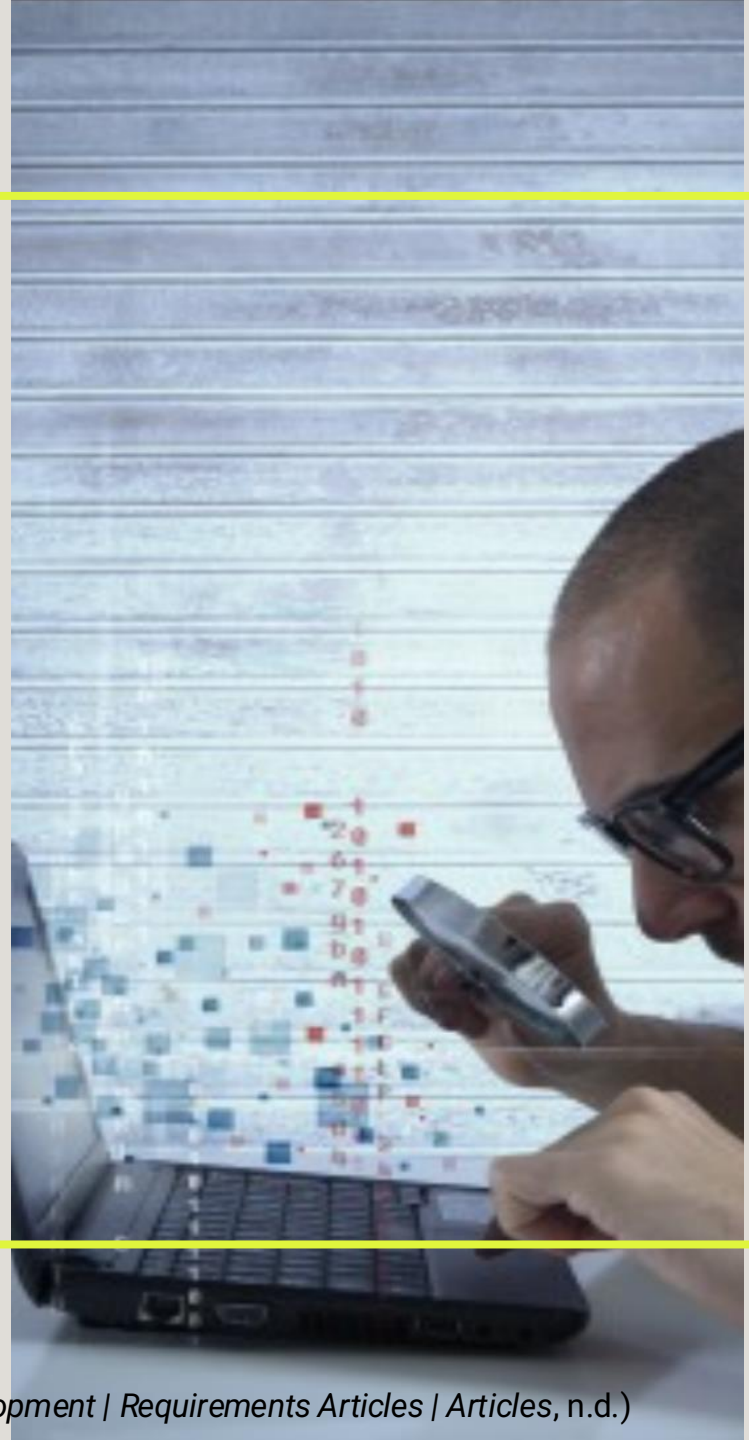
PROBLEM SOLVED BY THE PROJECT

Common banking issues addressed:

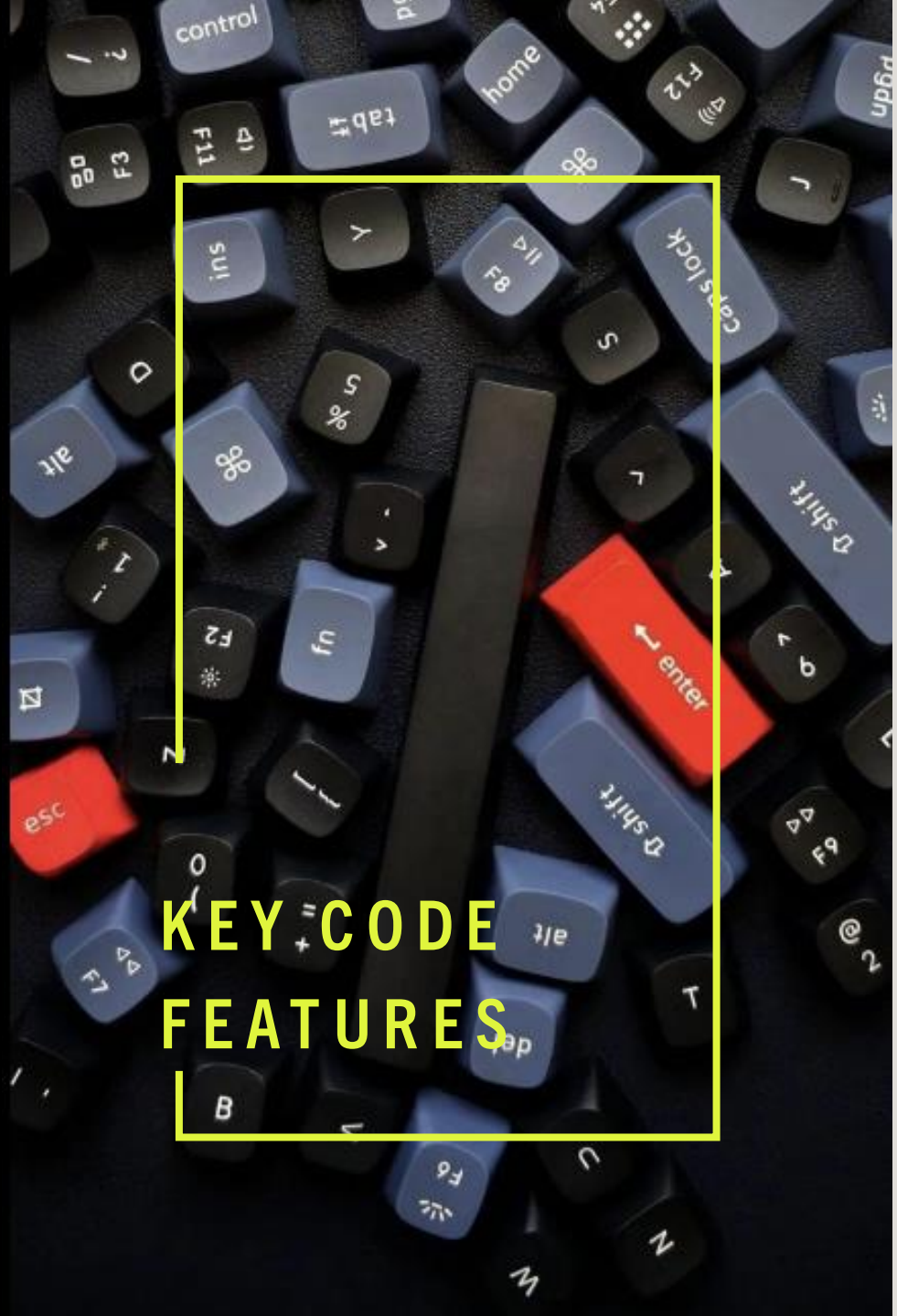
- Secure user authentication
- Prevention of invalid data entries
- Real-time transaction updates

Objectives achieved:

- Secure signup and login
- Error handling for invalid inputs
- User-friendly design



(*Solving the Right Problem – Step One in Software Development | Requirements Articles | Articles*, n.d.)



Signup:

Validates user details (username, email, age, phone, password).

Login:

Verifies credentials with limited attempts (max 3).

Deposit/Withdraw:

Updates balance securely with error handling for invalid inputs.

(Apple Scoop, 2024)

(*What Are the Seven Essential Features of a CAD Software?*, 2021)

Feature 1:

Improved error handling for invalid inputs.

Feature 2:

Added balance view functionality after every transaction.

TESTING THE APPLICATION

Content:

Test Case 1: Successful signup and login.

Test Case 2: Invalid login attempts (exceeding 3 attempts).

Test Case 3: Valid deposit and withdrawal.

Test Case 4: Invalid withdrawal (exceeding balance).

Test Case 5: Non-numerical input for transactions.

(*Software Application Testing Process: How to Test a Software Application?*, 2024)

TESTING THE APPLICATION

Successful signup and login

```
Welcome to the Bank!
1. Sign Up
2. Log In
3. Quit
Choose an option: 2

--- Log In ---
Enter Username: kira
Enter Password: 123456
Welcome, kira!
```

```
Welcome to the Bank!
1. Sign Up
2. Log In
3. Quit
Choose an option: 1

--- Sign Up ---
Enter Username: kira
Enter Email: kira@mail.com
Enter Age: 22
Enter Phone: 0222222222
Enter Password: 123456
Sign-up successful! Please log in to continue.
```

Invalid login attempts (exceeding 3 attempts).

```
Welcome to the Bank!
1. Sign Up
2. Log In
3. Quit
Choose an option: 2

--- Log In ---
Enter Username: 1
Enter Password: 1
Error: Invalid username or password. Please try again.
Enter Username: 1
Enter Password: 1
Error: Invalid username or password. Please try again.
Enter Username: 1
Enter Password: 1
Error: Invalid username or password. Please try again.
Error: Too many failed attempts.
Do you want to try again? (y/n): y
Enter Username: kira
Enter Password: 123123
Error: Invalid username or password. Please try again.
Enter Username: Joe.Doe
Enter Password: Password123
Welcome, Joe.Doe!

--- Banking Operations ---
1. Deposit
2. Withdraw
3. View Balance
4. Update Profile
5. Log Out
Choose an option:
```

Valid deposit and withdrawal

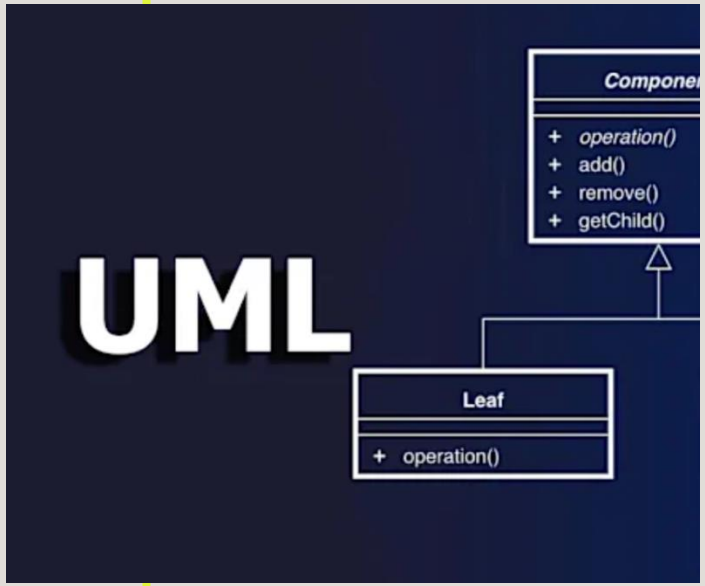
```
--- Banking Operations ---
1. Deposit
2. Withdraw
3. View Balance
4. Update Profile
5. Log Out
Choose an option: 1
Enter amount to deposit: 100
Successfully deposited $100. Current balance: $100

--- Banking Operations ---
1. Deposit
2. Withdraw
3. View Balance
4. Update Profile
5. Log Out
Choose an option: 2
Enter amount to withdraw: 20
Successfully withdrew $20. Current balance: $80
```

Non-numerical input for transactions

```
--- Banking Operations ---
1. Deposit
2. Withdraw
3. View Balance
4. Update Profile
5. Log Out
Choose an option: 2
Enter amount to withdraw: dd
Error: Invalid amount entered. Please enter a positive numeric value.
```

Invalid withdrawal (exceeding balance).

```
--- Banking Operations ---
1. Deposit
2. Withdraw
3. View Balance
4. Update Profile
5. Log Out
Choose an option: 2
Enter amount to withdraw: 100
Error: Insufficient funds.
```



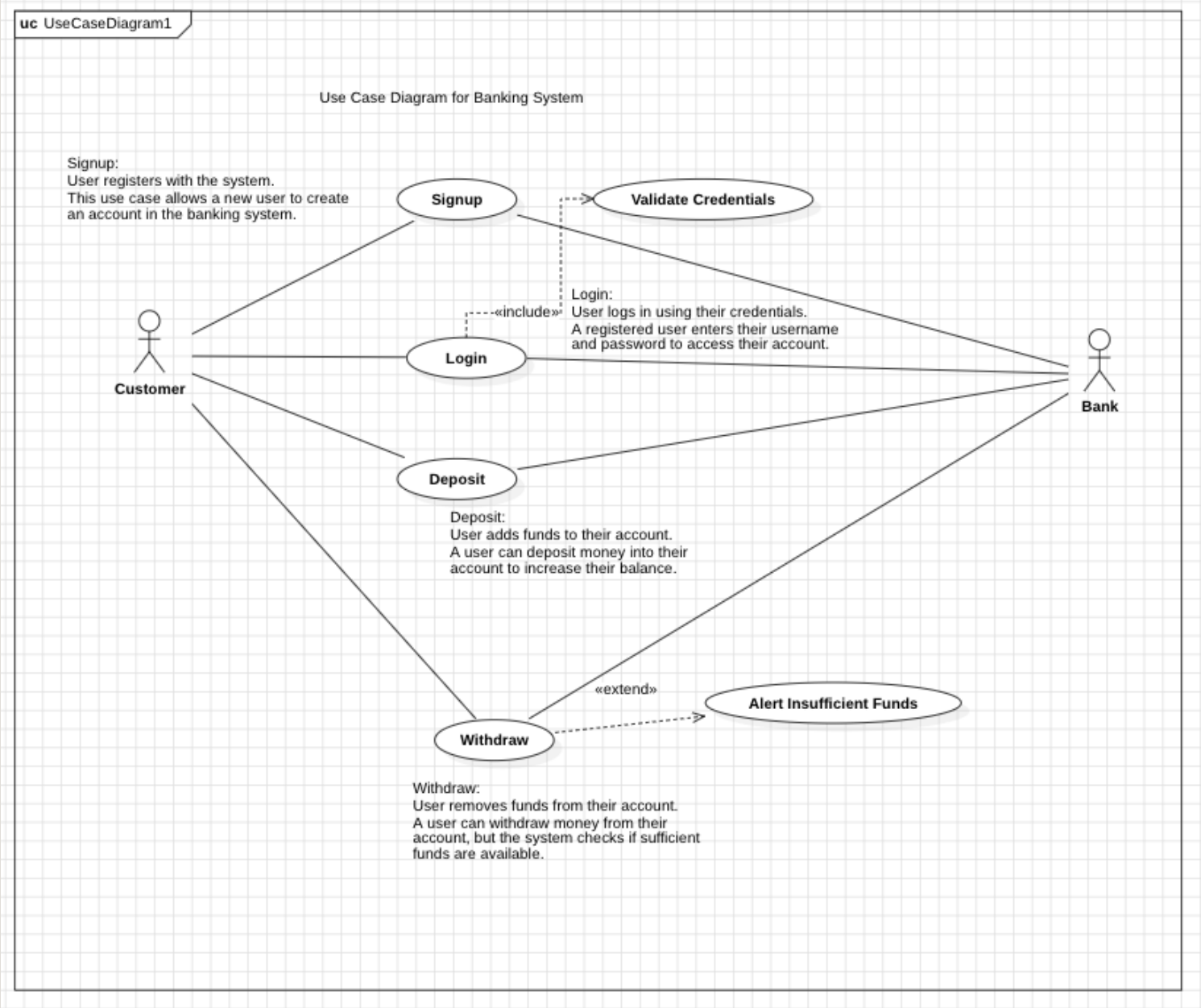
(*Erstellen Sie UML-Diagramme Für Ihre Projekte*, 2024)

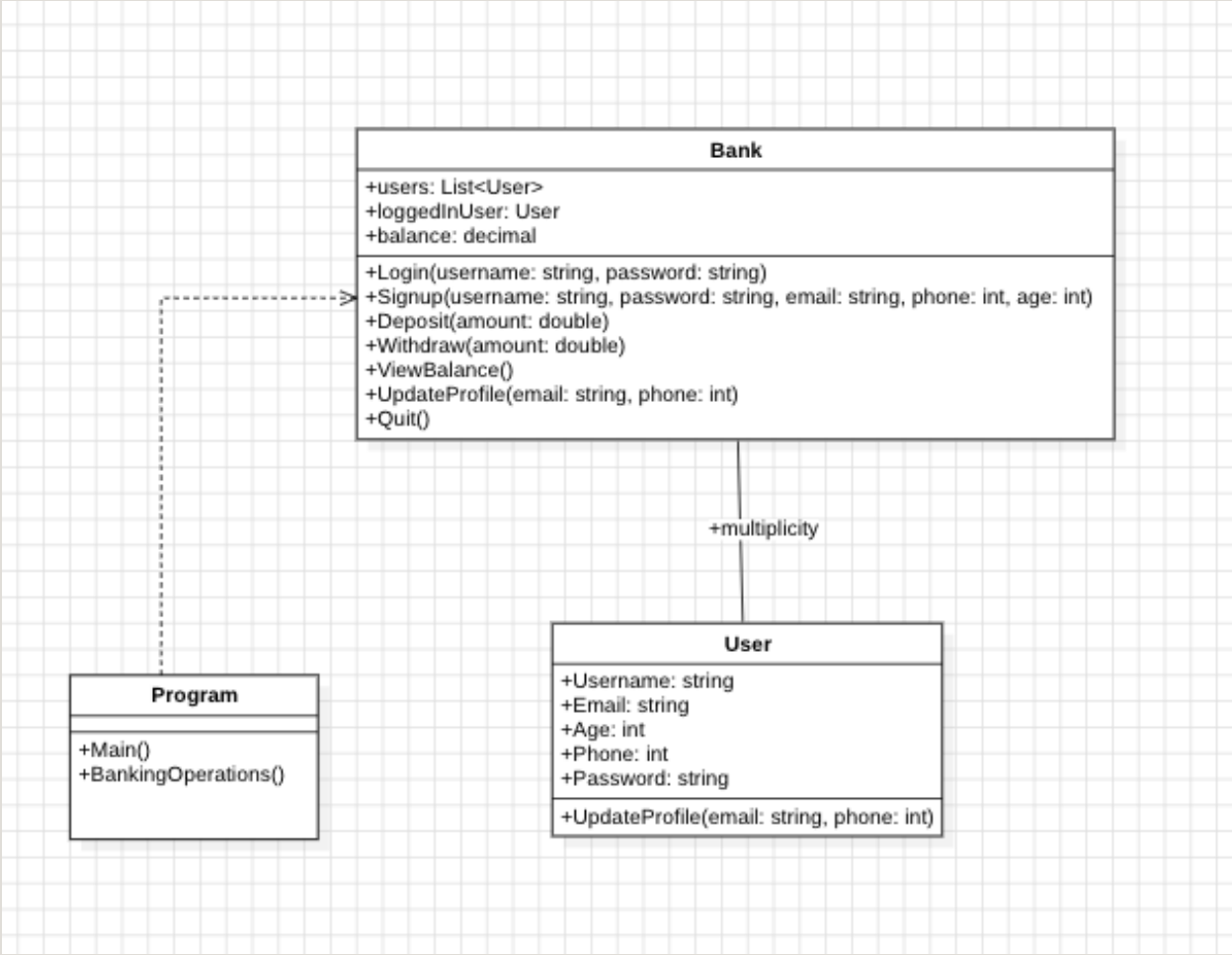
UML DIAGRAMS FOR BANKING SYSTEM

Content

- Use case diagram for all functionalities.
- Class diagram for core modules (Bank, User, Transaction).

USE CASE DIAGRAM FOR ALL FUNCTIONALITIES





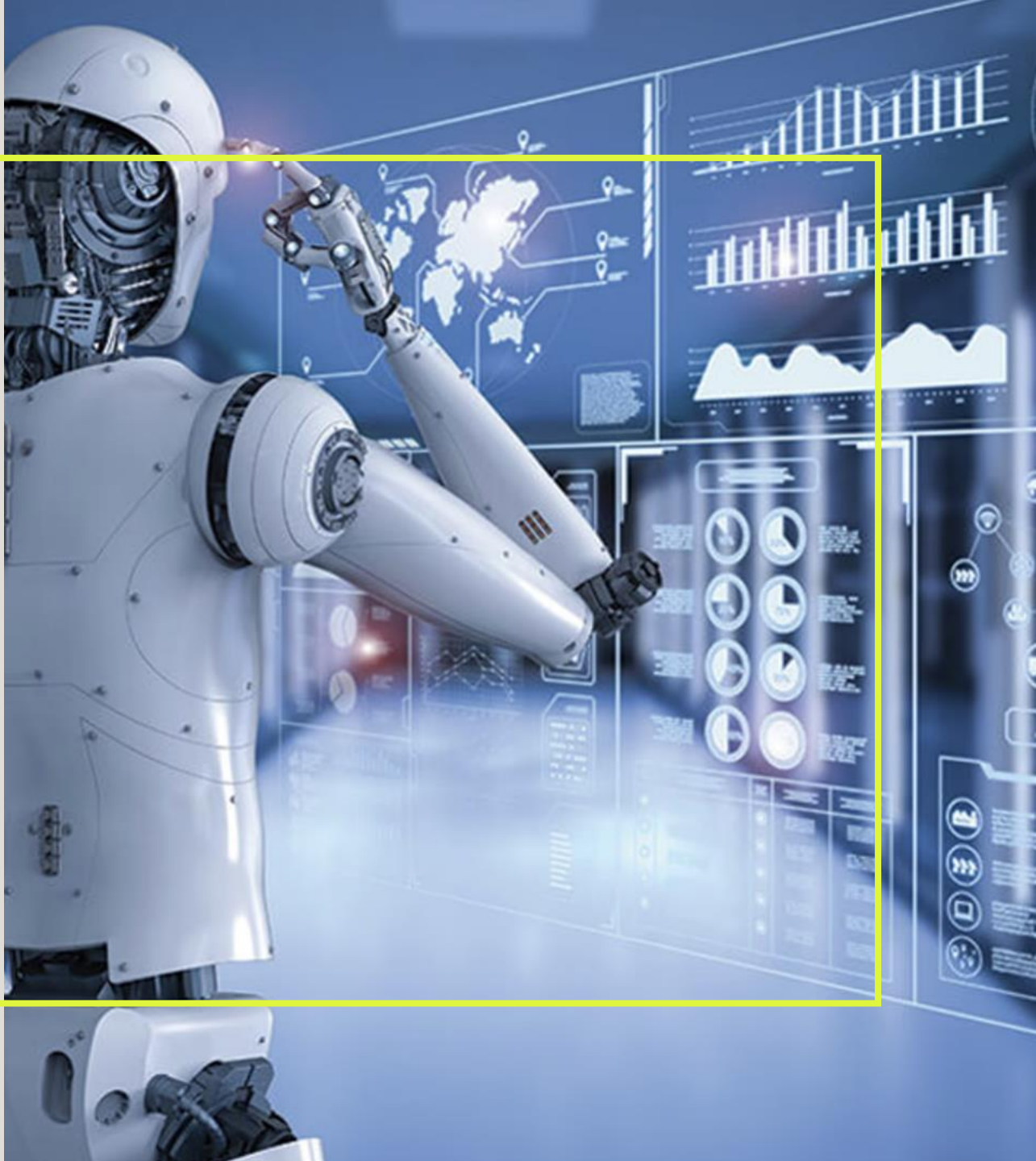
CLASS DIAGRAM FOR CORE MODULES (BANK, USER, TRANSACTION)

CONCLUSION AND FUTURE ENHANCEMENTS

Summary of accomplishments.

Future enhancements:

- Multi-user support
- Enhanced UI/UX



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THANK YOU

(*C Sharp, c Code, c Programmer, Code, Developers, HD Phone Wallpaper | Peakpx*, 2020)